



Is morphological priming stronger for transparent than opaque words? It depends on individual differences in spelling and vocabulary



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ABSTRACT

This experiment used the masked priming lexical decision task to address previous contradictory evidence about the relative strength of priming for (i) transparent pairs (e.g., *worker WORK*) which are morphologically and semantically related; (ii) opaque pairs (e.g., *corner CORN*) which appear to be morphological relatives but are not semantically related; and (iii) form pairs (e.g., *turnip TURN*) that are only orthographically related. The average data for 92 university students showed stronger priming effects for transparent than opaque or form pairs, due to a constant 'headstart' for related relative to unrelated pairs across the RT distribution. However, these average effects were significantly modulated by individual differences in independent measures of spelling and vocabulary. A 'semantic profile', defined by relatively higher vocabulary than spelling, was associated with robust priming for transparent pairs, particularly for slower responses, but little priming for opaque or form pairs. In contrast, individuals with an 'orthographic profile' of relatively higher spelling than vocabulary showed sustained priming for opaque pairs that was at least as strong as for transparently related pairs. This evidence of systematic individual differences amongst skilled readers has important implications for theories of lexical representation and processing.

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Introduction

Understanding how skilled readers process morphologically complex words (e.g., *trusted*, *distrust*, *untrustworthy*) provides insight into the organization and retrieval of lexical knowledge. As the smallest meaning-bearing linguistic unit, morphemes have the potential to serve as the elementary building blocks of word representations supporting an economical, non-redundant body of lexical knowledge that facilitates the learning of novel forms (e.g., *ungoogleable*) and morphological variants of known words (e.g., Merckx, Rastle, & Davis, 2011; Rastle & Davis, 2008; Taft & Forster, 1975). However, the utility of morphemes as linguistic building blocks has been challenged

by theoretical linguists (e.g., Hockett, 1987). The psycholinguistic validity of morphemic constituents is also questioned by computational models that effectively simulate apparent effects of morphological structure without assuming any explicit representation of morphological knowledge (e.g., Baayen, Milin, Filipovic-Durdevic, Hendrix, & Marelli, 2011; Plaut & Gonnerman, 2000). The role of morphological knowledge in lexical representation and processing therefore remains a focus of debate (e.g., Davis & Rastle, 2010; Feldman, O'Connor, & Moscoso del Prado Martín, 2009).

The masked priming task (Forster & Davis, 1984) has been one of the major sources of recent evidence about the role of morphological processing in word identification because it taps early lexical retrieval processes with little contamination from decision or response strategies associated with conscious awareness of the prime (see Kinoshita

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& Lupker, 2003, for reviews). Studies of masked priming between morphologically related words (e.g., *worker*–*WORK*; *darkness*–*DARK*) have consistently shown that morphologically complex words facilitate lexical decision responses to their stem morphemes across a range of languages (see Diependaele, Grainger, & Sandra, 2012; Rastle & Davis, 2008, for reviews) implying that “the morpheme-sized representations (shared by primes and targets) are automatically activated at a very early stage of visual word recognition” (Diependaele, Duñabeitia, Morris, & Keuleers, 2011, p. 344).

A central theoretical question underpinning research on morphological priming effects concerns whether they reflect abstract morphological representations or dynamic interactions between orthographic and semantic information during the early stages of lexical retrieval. The primary source of empirical evidence on this issue has come from comparisons of masked priming for ‘transparent’ and ‘opaque’ word pairs. Transparent pairs are genuine morphological relatives, both inflected (e.g., *worked*–*work*) and derived (e.g., *departure*–*depart*) forms, in which there is an explicit semantic relationship between the complex form and its stem. Opaque pairs share an apparent orthographic stem but are not semantically related. This includes complex forms which may have originally derived from the same stem but have acquired an idiosyncratic meaning (e.g., *department*–*depart*) and pseudomorphemic pairs in which the complex form contains a common affix which, when removed, yields another semantically unrelated word (e.g., *corner*–*corn*). Opaque pairs yield robust masked priming relative to equally orthographically similar ‘form primes’ that do not have an apparent morphological structure (e.g., *turnip*–*turn*). Critically, a number of studies have found that the magnitude of the priming effect for opaque pairs is just as large as that for transparent pairs (e.g., Longtin, Segui, & Halle, 2003; Rastle, Davis, & New, 2004). To confirm the generality of this finding, Rastle and Davis (2008) collated the results of 19 published studies assessing morphological priming using brief (<60 ms) masked primes, which revealed that the average priming effect for transparent pairs (30 ms) was only slightly larger than that for opaque pairs (23 ms) and both effects were substantially stronger than for orthographically matched form primes (2 ms).

This evidence has been taken as support for ‘morphographic decomposition’ accounts of morphological processing (Rastle & Davis, 2008; Rastle et al., 2004) which assume an early orthographically-driven segmentation process that parses complex words into affixes and stems to allow access to a morphologically structured lexicon. The first version of this proposal adopted a serial search framework (Taft & Forster, 1975) in which a preliminary ‘affix-stripping’ process isolated the stem of the word required to search through the lexicon, which was organized by morphological stems. This decomposition account was subsequently reconceptualised within an interactive activation (IA) framework (Taft, 1994) in which frequently occurring affixes like *re-*, *-ed* and *-ing* are represented as sublexical units that can be activated in the early stages of lexical retrieval.

Such views have been dubbed ‘form first’ (Forster & Hector, 2002) or ‘form-then-meaning’ accounts (Feldman et al., 2009) because they assume that “recognition of morphologically complex words begins with a rapid morphemic segmentation based solely on the analysis of orthography” (Rastle & Davis, 2008, p. 942). Within these frameworks, the semantic information responsible for ‘transparency effects’ (i.e., stronger priming for transparent than opaque pairs) is attributed to top-down semantic feedback that occurs later in processing, or to some form of independent ‘morpho-semantic’ decomposition process (Rastle & Davis, 2008) perhaps involving a checking or ‘licensing’ (Schreuder & Baayen, 1995) procedure to evaluate the legality of stem-affix combinations.

Most other models of morphological representation and processing predict a graded effect of morphological structure whereby priming for transparent pairs (e.g., *worker*–*work*) is greater than for opaque pairs (e.g., *corner*–*corn*) which, in turn, produce more priming than orthographic controls (e.g., *turnip*–*turn*). Different models account for the early semantic influences on priming in different ways.

‘Dual route’ models assume that written words are mapped, in parallel, to morphemic and whole word representations (e.g. Diependaele, Sandra, & Grainger, 2005). Transparency effects are predicted by such models because activation from the morphemic and whole word routes converge on related semantic units (e.g., those associated with *worker* and *work*), while opaque words like *corner* and *corn* do not share semantic features (Diependaele et al., 2011). Similar predictions derive from Taft and Nguyen-Hoan’s (2010) recent extension of the IA framework which adds a layer of ‘lemma’ representations between the form and semantic levels, at which lemmas for transparent, but not opaque words, benefit from facilitative connections between related words.

Graded effects emerge most naturally in parallel distributed processing (PDP) models of morphological processing (e.g., Plaut & Gonnerman, 2000) in which statistical learning mechanisms capture the systematic correlations between form and meaning that define morphology. Within these models there are no discrete representational structures corresponding to either words or morphemes. Systematic covariations between orthography, phonology and semantics shape the patterns of activation generated among the hidden units that connect these networks so that words that share both form and meaning become associated with more similar hidden unit representations (e.g., Plaut & Gonnerman, 2000). Rather than reflecting an independent level of representational structure, morphological relationships are therefore an emergent consequence of correlations between form and meaning (Seidenberg & Gonnerman, 2000). Such models clearly predict that transparent pairs, which share both orthographic and semantic features, will yield stronger priming than opaque pairs that share only orthography. It is less clear that PDP models should yield stronger priming for opaque pairs (e.g., *corner*–*corn*) than orthographically-matched controls (e.g., *turnip*–*turn*) since neither are semantically related. As elaborated in the General Discussion, Plaut and Gonnerman (2000) have shown that this may depend on the ‘morphological richness’ of the language.

Statistical learning also underlies the morphological effects that emerge in Baayen et al.'s (2011) recent Naïve Discriminative Learning (NDL) model of morphological processing. Like PDP models, despite containing no explicit representation of morphemes, this framework captures sensitivity to morphological structure through learned associations between orthography, represented by letter and bigram units, and meaning, represented as symbolic localist codes for root morphemes and grammatical functions, but not for morphologically complex word forms. This two-layer network generally produces graded effects of morphological structure that are a function of the morphological family size of the stem morpheme and the informativeness of the affix. However, Baayen et al. (2011) showed that a variant of the NDL model accurately simulated the significant priming obtained for Opaque pairs reported by Rastle et al. (2004) without assuming an explicit morphographic parsing process.

Thus, many models of morphological processing predict stronger priming for transparent than opaque word pairs. Empirical demonstrations that they produce equivalent priming have therefore been a focus of considerable scrutiny. In contrast to the conclusion of Rastle and Davis's (2008) qualitative review, a quantitative meta-analysis of 16 studies comparing masked priming for transparent and opaque words showed significantly stronger priming effect for transparent than opaque words, even though the effect was not significant in a number of the individual studies (Feldman et al., 2009). Feldman et al. (2009) also reported new empirical data comparing large sets of transparent and opaque pairs matched on both stem and affix frequency which confirmed previous studies in other languages (e.g., Diependaele et al., 2005) by showing significant transparency effects from 50 ms masked primes under conditions in which opaque primes yielded a non-significant priming effect of only 4 ms relative to orthographically unrelated primes.

However, it remains difficult to definitively refute the form-first view. In addition to the ambiguities in clearly operationalizing the distinction between transparent and opaque words and ensuring that they are matched on all relevant variables (Davis & Rastle, 2010), the hierarchical IA-style framework proposed by Rastle et al. (2004) assumes cascaded processing. This view predicts only that there is "a moment in time where morphemic activations are *purely* morpho-orthographic in nature" (Diependaele et al., 2011, p. 354). Definitively demonstrating that there are *no* semantic effects at very early stages of processing therefore raises the range of statistical issues associated with proving a null hypothesis (Cribbie, Gruman, & Arpin-Cribbie, 2004). To shed light on these theoretical and empirical ambiguities, the present research introduces a new approach that uses individual differences in spelling and vocabulary to disentangle orthographic and semantic contributions to morphological priming.

Individual differences in masked morphological priming

All of the data reviewed above has been based on the average results of samples of university students selected,

at most, for being native speakers of the relevant language. Like the bulk of the experimental psycholinguistic literature, this reliance on average data reflects an implicit 'uniformity assumption' (Andrews, 2012): that skilled reading is characterized by a uniform, cognitive architecture that can be inferred from average RT data for samples of 20–30 student participants. Although this assumption has served the field well in establishing the general principles of skilled lexical processing (e.g., Balota, Cortese, Segent-Marshall, Spieler, & Yap, 2004), Andrews (2012) argues that further theoretical advances require consideration of individual differences amongst skilled readers, particularly in domains like morphological priming where the empirical results obtained from averaged data have proved inconclusive.

The present research investigated whether and how individual differences in orthographic processing (indexed by spelling) and semantic processing (indexed by vocabulary) predicted priming for transparent and opaque word pairs in a large sample of native English speaking university students. The selection of vocabulary and spelling as individual difference measures was based on two recent bodies of evidence showing systematic individual differences amongst skilled university student readers in both semantic priming (Yap, Tse, & Balota, 2009) and orthographic priming (Andrews & Hersch, 2010; Andrews & Lo, 2012).

Yap et al. (2009) compared semantic priming for high and low frequency words in groups of participants that differed in their average vocabulary score. Consistent with the typical result reported in average data for unselected student samples (e.g., Borowsky & Besner, 1993; Plaut & Booth, 2000), the lower vocabulary group showed a significant interaction between priming and frequency due to larger semantic priming effects for low than high frequency words. However, the higher vocabulary group showed additive effects of word frequency and semantic priming.

To further probe the differences between higher and lower vocabulary readers, Yap et al. (2009) analyzed the reaction time (RT) distributions for the two groups, which revealed that the interaction between frequency and semantic priming observed for lower vocabulary readers was primarily due to a much larger benefit from semantic primes in the slow tail of the RT distribution – typically, responses to more difficult, low frequency words. In contrast, the higher vocabulary group showed a small, consistent benefit from semantic primes across the entire RT distribution. Yap et al. (2009) argued that the general shift in the central tendency of the RT distribution shown by higher vocabulary participants is consistent with an automatic, modular lexical system that allows primes to "provide a headstart that is independent of the target's difficulty" (p. 321) whereas the increased priming for slow responses shown by low vocabulary participants reflects "greater reliance on prime information for resolving these difficult targets, with reliance being proportional to the difficulty of the trial" (p. 317). They attributed the differences between higher and lower vocabulary participants to individual differences in the 'integrity' or 'quality' of their lexical representations.

Lexical quality is a construct proposed by Perfetti (1992) to capture variability in the strength and precision of lexical representations both between competent readers, and within the written word vocabulary of an individual skilled reader. High quality lexical representations are defined as precise, redundant and coherent (Perfetti, 2007): they precisely specify a word's orthography and its redundant links to phonology and support coherent activation of the orthographic, phonological and semantic information that define the identity of a word. Yap et al. (2009) interpreted their findings as showing that high vocabulary scores index high quality lexical representations that support fast, fluent lexical processing of the prime which confers a small benefit for semantically related targets, even for low frequency words. However, low vocabulary participants who have poorer quality lexical representations, particularly for lower frequency words, need to use the semantic relationships between primes and targets to facilitate identification of difficult words, in the same way that average readers use semantic information to identify more difficult words when they are presented in a degraded format (Balota, Yap, Cortese, & Watson, 2008). Thus, Yap et al.'s (2009) data demonstrate that average priming effects for samples of skilled readers can obscure systematic differences in semantic priming that are predicted by measures of vocabulary. These findings are consistent with the lexical quality account but, as Yap et al. (2009) acknowledge, vocabulary is a 'crude proxy' for the construct of lexical quality.

Lexical quality also underpinned our own recent investigations of individual differences in masked orthographic priming paradigm (Andrews & Hersch, 2010; Andrews & Lo, 2012), which used a battery of measures of written language proficiency. In addition to measuring vocabulary, reading comprehension and reading speed, we also assessed spelling ability to capture the orthographic precision that is central to Perfetti's (1992) definition of high quality lexical representations. Despite the strong correlations between these measures, spelling has consistently proved to be the most potent unique predictor of individual differences in masked orthographic priming, apparently because it selectively taps individual differences in sensitivity to lexical competition.

This was first demonstrated in Andrews and Hersch's (2010) comparison of individual differences in masked neighbor priming (e.g., note NODE) for words with many or few lexical neighbors (High N vs Low N words¹). The typical finding in average data is that neighbor priming only occurs for Low N words (Forster, Davis, Schoknecht, & Carter, 1987), leading to claims that the lexical representations for High N words are 'recoded' (Forster & Taft, 1994) or 'tuned' (Castles, Davis, Cavalot, & Forster, 2007) to differentiate them from their many orthographically similar neighbors. However, we found that the null priming effect for High N words observed in mean RT data for the total sample arose from averaging opposite priming effects for better and poorer spellers: poor spellers showed facilitative neighbor

priming while better spellers showed inhibitory priming (Andrews & Hersch, 2010). This differential priming was specific to High N words with many neighbors; better and poorer spellers both showed facilitative priming for Low N words. A second experiment replicated the finding of stronger inhibitory neighbor priming in better spellers and also showed that only better spellers demonstrated reduced priming from Ambiguous partial primes (e.g., *crow#CROWN*), which could potentially activate more than one word (e.g., *crown*, *crowd*), relative to Unambiguous partial primes (e.g., *cr#wn CROWN*) in which the target is the only word that is consistent with the prime.

These results were interpreted as showing that better spelling is selectively associated with lexical competition: better spellers showed inhibitory priming for High N words and reduced facilitation for partial primes that activated more than one word, but for Low N targets and Unambiguous partial primes, they showed facilitative priming effects just as strong as those shown by poorer spellers. The selective effects of spelling are consistent with Perry, Lupker and Davis's (2008) claim, derived from computational simulations of the IA model, that masked neighbor priming effects arise from the combined effects of sublexical facilitation and lexical competition. Only the latter appears to be sensitive to spelling.

To further explore this possibility, Andrews and Lo (2012) compared prime lexicality effects for neighbor and transposed-letter (TL) primes. Confirming the independent contributions of sublexical facilitation and lexical competition, we found that higher general proficiency, indexed by a principal component combining reading, spelling and vocabulary, was associated with both stronger inhibition from orthographically similar word primes, and stronger facilitation from orthographically similar nonword primes. Higher general proficiency was also associated with stronger inhibition from TL primes (e.g., *colt CLOT*), which share all their letters with the prime, than from neighbor primes (e.g., *plot CLOT*), which differ in one letter. However, consistent with Andrews and Hersch's (2010) data, spelling was selectively related to lexical competition as revealed by a second principal component that differentiated spelling from reading and vocabulary. Individuals whose spelling ability was higher than expected for their reading and vocabulary showed inhibitory effects of orthographic primes, particularly TL primes, while relatively poor spelling was associated with facilitative priming.

The present study

The present study investigated whether the modulating role of individual differences among skilled readers on masked priming extends to morphological priming effects. More specifically, the goal was to evaluate orthographic and semantic contributions to morphological priming by assessing whether spelling and vocabulary exert independent effects on masked priming for opaque and transparently related word pairs. On the basis of our previous evidence that spelling indexes the orthographic precision of lexical representations (Andrews & Hersch, 2010; Andrews & Lo, 2012), we anticipated that accurate spelling would be associated with a 'morphographic' pattern of

¹ Most studies have defined neighbors using the metric developed by Coltheart et al. (1977) which counts words of the same length as a target word that differ by only one letter.

priming defined by equivalent priming for opaque and transparent pairs. If vocabulary scores capture individual differences in semantic coherence (Perfetti, 2007) or 'lexical integrity' (Yap et al., 2009), vocabulary may predict the semantic influences that yield the additional boost in priming for transparent relative to opaque pairs. Following Yap et al. (2009) we also used RT distribution analysis to assess the time course of morphological priming and how it is modulated by individual differences in spelling and vocabulary.

Method

Participants

The participants were 100 undergraduate students (78 female; average age 19.1) who participated in exchange for course credit. All had begun speaking English by age five.

Measures of individual differences

Participants were assessed on three measures of written language proficiency that are described in detail by Andrews and Hersch (2010): a test of vocabulary to index semantic knowledge; and two tests of spelling, a dictation test and a spelling recognition test. Andrews and Hersch (2010) reported that the test–retest reliabilities of these instruments ranged from 0.86 for Vocabulary to 0.93 for Spelling Recognition. Each of the individual measures was standardized, and a composite measure of spelling (ZSpell) was calculated by averaging the standard scores of the two spelling tests, which were highly correlated ($r = .82$).

Experimental design and stimuli

The design and examples of each condition are presented in Table 1. Ninety prime–target pairs were selected from the items used by Rastle et al. (2004), 30 for each of three conditions. Primes for the Transparent and Opaque conditions could be parsed into the target and an English suffix and were pairwise-matched on their suffix. Transparent pairs contained an orthographically regular, semantically transparent morphological relationship between the prime and target (e.g., *dreamer DREAM*). Opaque pairs were selected that appeared to share the same morphological relationship as their matched transparent pair but the complex prime bore no semantic relationship to the target pseudo-stem² (e.g., *flicker-FLICK*). Primes in the Form condition began with the target word but had a non-morpholog-

Table 1

Examples of prime and target words and average frequency and length for each condition.

	Target	Primes	
		Related	Unrelated
Transparent	TEACH	Teacher	Finally
Log frequency	1.28	0.85	0.88
Length	5.00	7.30	7.27
Opaque	COAST	Coaster	Muffler
Log frequency	0.94	0.90	0.92
Length	4.87	7.17	7.20
Form	PULP	Pulpit	Gifted
Log frequency	0.96	0.73	0.89
Length	4.47	6.73	6.73

ical ending (i.e. a letter sequence that is not used as a suffix in English) so, though they were just as orthographically similar to the target as the Transparent and Opaque pairs, there was no semantic or apparent morphological relationship between the prime and target (e.g. *pulpit-PULP*). The unrelated control primes used by Rastle et al. (2004) were retained for each of the 90 target words selected from their stimulus set. Control primes were orthographically, morphologically, and semantically unrelated to their target word and were matched as closely as possible to each related prime on frequency and length (see for Appendix A a complete list of word stimuli).

The critical stimulus set also included 90 legal non-words created using the ARC nonword database (Rastle, Harrington, & Coltheart, 2002) to match the target words in length and number of orthographic neighbors. Each target nonword was paired with either an orthographically related or unrelated nonword prime. Orthographically related primes were created by applying the same suffix or affix to the nonword target as their matched word trials. The stimulus list included a further 160 filler pairs (80 word and 80 nonword targets), half of which were preceded by orthographically related primes. To avoid exposing participants to repeated targets within a list while collecting data for all targets in all priming conditions, six stimulus lists were constructed that rotated prime–target combinations across the different conditions.

Procedure

Each trial in the masked priming lexical decision task consisted of three successive displays presented in the center of a computer screen in white 12 pt Courier font on a black background: a forward mask (#####) presented for 500 ms, the prime stimulus presented in lower case for 50 ms, and the target word in upper case for 500 ms. The DMDX display software (Forster & Forster, 2003) was used to control stimulus presentation and to record reaction time (RT) and errors for each trial. The software generated a different individual order of items for each participant. Participants were instructed that they would be presented with stimuli that they were required to classify as words or nonwords. No mention was made of the primes. Participants responded by pressing one button, using their dominant hand, if the stimulus was a word and another

² It is difficult to unambiguously distinguish Transparent and Opaque pairs. For example, the meaning of the pseudo-stem *earl* is not related to the complex words *early*, but as Baayen et al. (2011) note, it is historically related to the stem *ere* and the suffix *-ly* remains a functional adverbial marker. Some words also have multiple meanings that vary on this dimension: as well as meaning jovial and vigorous, *hearty* is associated with 'heartfelt' and in some dialects *coaster* refers to both a coastal ship and a protective mat placed under a glass. Since our goal was to investigate whether individual differences modulate previously reported differences between transparent and opaque pairs, we maintained the same classifications as the widely used Rastle et al. (2004) stimulus set.

button, using their other hand if it was a nonword. The experimental list was preceded by 16 practice trials.

Results and discussion

Five targets with overall error rates of over 50% were excluded from all analyses. Three participants who made more than 10% anticipatory responses and/or 25% errors, along with one extreme outlier in the linear mixed analyses and four participants whose data produced unreliable RT quantile estimates were excluded from all analyses. The data for the remaining 92 participants were cleaned to remove incorrect responses (8.4% of total), as well as anticipatory RTs of less than 200 ms and outliers of greater than 1500 ms (0.1% of total).

ANOVA analyses

Table 2 presents the mean RT and accuracy for each word target condition, averaged across the whole sample. To allow comparison with previous reports of average data for student samples, conventional 2 (prime type) $\times$ 3 (target type) ANOVAs were first conducted separately on the subject (F_1) and item-wise (F_2) data for the complete sample. Both factors were within groups for the subject-wise analysis, while target type was between groups for the item-wise analysis. List (6 levels) was also included as a between groups factor in both subject and item-wise analyses to remove variance due to the counterbalancing procedure (Pollatsek & Well, 1995). The results revealed a significant main effect of prime type ($F_1(1,84) = 45.775$, $p < .001$; $F_2(1,79) = 31.680$, $p < .001$), because related pairs were responded to more quickly than unrelated pairs. The effect of target type was also significant in the subjects-wise analysis, because form stems were responded to more quickly than opaque ($F_1(1,84) = 4.524$, $p = .036$; $F_2(1,79) = 0.860$, $p = .357$) or transparent stems ($F_1(1,84) = 16.601$, $p < .001$; $F_2(1,79) = 1.019$, $p = .316$), but RTs for opaque and transparent stems did not significantly differ ($F_1(1,84) = 2.404$, $p = .125$; $F_2(1,79) = 0.009$, $p = .926$). Importantly, the facilitative priming effect was significantly modulated by target type ($F_1(2,168) = 4.439$, $p = .013$; $F_2(2,79) = 3.670$, $p = .030$). Planned contrasts revealed significantly greater facilitation for transparent pairs than form pairs ($F_1(1,84) = 10.113$, $p = .002$; $F_2(1,79) = 7.051$, $p = .01$), but no significant differences between opaque and form ($F_1(1,84) = 1.235$, $p = .270$; $F_2(1,79) = 0.914$, $p = .342$), or transparent and opaque pairs ($F_1(1,84) = 3.180$, $p = .078$; $F_2(1,79) = 3.135$, $p = .080$). General linear models including the individual difference variables as covariates revealed significant interactions that were explored in the mixed linear analyses reported below.

Table 2

Average RT (and percentage error rate) for each condition across the whole sample.

Prime type	Target type		
	Form	Opaque	Transparent
Unrelated	589 (10.5)	586 (8.5)	587 (8.7)
Related	579 (10.0)	567 (6.5)	554 (7.0)
Priming	10 (0.5)	19 (2.0)	33 (1.7)

ANOVA analyses of the accuracy data showed that fewer errors were made to related than unrelated pairs ($F_1(1,84) = 7.791$, $p = .006$; $F_2(1,79) = 3.598$, $p = .062$). However, there were no significant interactions with target type ($F_1(2,168) = 0.592$, $p = .554$; $F_2(2,79) = 0.380$, $p = .685$) or with the individual difference covariates, so accuracy was not analyzed further.

Despite our large sample size, the ANOVA outcomes parallel the statistically ambiguous findings that pervade the literature. RT priming for transparent pairs was significantly stronger than for form controls but not opaque pairs, but opaque pairs did not differ significantly from form pairs either. Although the stimuli were carefully matched, this analysis also revealed differences in average RT for the different stem conditions, independent of prime type. However, these effects were also statistically ambiguous because they reached significance when averaged over subjects but not items, raising questions about their generality. Mixed linear analysis methods modeling address many of these issues (see Baayen, 2008; Baayen, Davison, & Bates, 2008).

Linear Mixed Effect (LME) analysis of word RT

LME models were tested on the RT data for word targets using the lme4 package (Bates, Maechler, & Dai, 2010). This analysis approach is ideally suited to the present research because it allows the effects of crossed and nested subject and item factors to be considered within a single analysis. We can therefore simultaneously assess generality over both participants and items while statistically controlling for a number of sources of extraneous variability. Previous applications of this approach have shown that, even when stimuli are carefully matched and counterbalanced, residual effects of stimulus attributes, list construction and trial sequence can explain substantial, systematic variance in RT, and that partialling out this variance can both eliminate artifactual effects and increase power to detect real effects (Baayen, 2008).

The analyses treated participants, targets and list as crossed and partially nested random effects³ and were conducted on inverse RT because the Box Cox procedure (1964) showed that this transformation provided the best approximation to normal-distribution assumptions. To assess the effects of target and prime conditions, normalized sum contrasts separately compared the opaque and transparent target types with the form condition. A generalized matrix inversion was then conducted on the contrast weights in order to orthogonalize the sets of contrasts and yield interpretable main effects (Venables & Ripley, 2002). Higher-order interactions of these contrasts with a similar, generalized matrix inverted contrast on prime type were also included as fixed effects. A second model tested generalized

³ Estimated variance in RT due to participants (inverse $SD = 0.17745$, raw $SD = 63.08$ ms), items (inverse $SD = 0.095$, raw $SD = 34.63$ ms), and lists (inverse $SD = 0.06353$, raw $SD = 22.50$ ms) was 27.9%, 15.0% and 10.0% of total variability in RT, respectively. According to nested χ^2 tests, model fit was significantly improved by including participants and items in the random effect structure, but not by including list. List variability was nonetheless retained in the model to allow non-trivial adjustments based on random list assignment.

matrix inverted contrasts that compared (i) the mean of opaque and transparent target stems with the form condition, and (ii) opaque with transparent target conditions. All continuous predictors and covariates were mean centered. Likelihood ratio tests assessed whether model fit was significantly improved by including by-subject or by-item variability in the effects of stimulus and task variables in the random structure of the model.⁴ The final models were also run using raw RT to provide estimates of the effects (Est. Raw RT in analyses below) that can be compared with previous literature.

The composite ZSpell measures and standardized Vocabulary scores (zVocab) were highly correlated ($r = .67$). To facilitate interpretation, two orthogonal measures of individual differences were derived from a principal components analysis. The first principal component (PC1) had high positive correlations with both spelling ($r = .91$) and vocab ($r = .91$) and captured 84% of the common variance between them, while the second principal component (PC2), accounting for the remaining 16% of variance, showed opposite directions of relationship with ZSpell ($r = .41$) and zVocab ($r = -.41$) and captured the difference between them.

The initial stage of analysis showed that, despite the careful matching and counterbalancing of stimuli, there were significant fixed effects of target neighborhood size ($t = -3.55$, $p\text{MCMC} < .001$, Est. Raw RT = -9 ms), log frequency of the target ($t = -7.1$, $p\text{MCMC} < .001$, Est. Raw RT = -51 ms), the number of trials the participant has completed ($t = -5.17$, $p\text{MCMC} < 0.001$, Est. Raw RT = -1 ms), RT on the participant's previous trial⁵ ($t = 8.18$, $p\text{MCMC} < .001$, Est. Raw RT = 0 ms), and congruency of the previous response

($t = -3.12$, $p\text{MCMC} = 0.002$, Est. Raw RT = -12 ms).⁶ Consistent with previous literature, more previous trials, a prior consistent motor response, higher target word frequency and target neighborhood size were all associated with faster classification while longer RT on the participant's previous trial predicted slower responses to the current target.

With these fixed and random effects controlled, the analysis showed a highly robust main effect of priming ($t = -4.92$, $p\text{MCMC} < 0.001$, Est. Raw RT = -21 ms) that significantly interacted with target type. There was significantly greater facilitative priming for transparent than form ($t = -3.6$, $p\text{MCMC} < 0.001$, Est. Raw RT = -26 ms) or opaque pairs ($t = -2.22$, $p\text{MCMC} = 0.016$, Est. Raw RT = -16 ms), but no difference in priming between form and opaque pairs ($t = -1.5$, $p\text{MCMC} = 0.114$, Est. Raw RT = -10 ms).

Thus, the overall effects revealed by the LME models showed a clear transparency effect. Although opaque pairs yielded significant priming, the effect was significantly smaller than that for transparent pairs, and did not differ significantly from orthographically matched form controls. These average data provide no support for the morphographic effects predicted by the form-first accounts of morphological priming.

The critical novel question addressed by this research is whether these average priming effects are modulated by individual differences. Fig. 1 presents the relationship between each of the principal components and RT to targets preceded by related and unrelated primes, separately for the form, opaque and transparent conditions. The upper panels show the relationship of RT to the first principal component, PC1, indexed by the sum of spelling and vocabulary, which tapped general proficiency. There was no significant main effect of PC1, ($t = -1.23$, $p\text{MCMC} = 0.148$), but the analysis revealed significant 2-way interactions between general proficiency and the difference between average RT for the transparent condition and both the form ($t = -1.99$, $p\text{MCMC} = 0.05$, Est. Raw RT = -5 ms) and opaque conditions ($t = -2.42$, $p\text{MCMC} = 0.02$, Est. Raw RT = -8 ms) which occurred because the association between higher proficiency and faster RT was significantly stronger for transparent target stems. However, PC1 did not significantly modulate any effects involving priming ($p\text{MCMC} > 0.20$).

Priming did, however, significantly interact with the second principal component, PC2, which reflected unique variance differentiating spelling ability from vocabulary with general proficiency partialled out. Higher values on this factor indicate that spelling scores were relatively higher than vocabulary, while lower values indicate the reverse relationship. As illustrated in the lower panels of Fig. 1, a significant 3-way interaction between PC2 and the magnitude of priming effects for opaque relative to transparent pairs ($t = 2.36$, $p\text{MCMC} = 0.02$, Est. Raw RT = 34 ms) arose because superior spelling relative to

⁴ The lmer syntax for the final model was: `inverse_RT ~ target.log_freq + target.Nsize + prime.log_freq + trial_num + prev_RT*prev_accuracy*prev_Resp_Congruency + (TargetType*PrimeType)*PC1 + (TargetType*PrimeType)*PC2 + (1|Subject) + (0 + prev_RT|Subject) + (0 + prev_Resp_Congruency|Subject) + (0 + trial_num|Subject) + (0 + target.log_freq|Subject) + (0 + prime.log_freq|Subject) + (0 + PrimeType|Subject) + (1|Target) + (0 + PC1|Target) + (0 + PC2|Target) + (0 + PrimeType|Target) + (1 + PrimeType|List)`. By-subject random slopes for target log frequency (raw SD: 19 ms), number of trials (raw SD = 0.13 ms), previous RT (raw SD = 0.10 ms) response congruency (raw SD = 24 ms) and prime log frequency (raw SD = 7.91 ms) were included in the random effect structure of the final model to take account of variability between participants in the effects of these stimulus and task variables. Although the magnitude of priming did not significantly differ across individual subjects (inverse SD = 0.03456, raw SD = 11 ms, $p = 0.15$), or individual lists (inverse SD = 0.03023, raw SD = 11 ms, $p = 0.12$), the correlation between these factors, which occurred because slower lists yielded significantly less priming ($r = 1$, $p = 0.02$), was included in the model. Model fit was also improved by allowing for the significant differential sensitivity between individual items to vary for both PC1 (inverse SD = 0.0224, raw SD = 8 ms) and PC2 (inverse SD = 0.02715, raw SD = 12 ms).

⁵ The robust effect of RT on the previous trial is consistent with the evidence of sequential effects in previous applications of LME analysis (e.g. Baayen, 2008). However, previous studies rarely specify exactly how this was operationalised. Critically, the definition of what constitutes a valid previous trial alters the number of datapoints for the statistical analysis. For example, adopting the criterion used by Kinoshita, Mozer, and Forster (2011) of only including correct trials for which the previous trial was also correct would have reduced the potential set of 7113 datapoints by 9.2%. We instead included previous RT for both correct and incorrect trials and only excluded aberrant datapoints (<200 and >1500 ms) using the same criteria as for current RT. The reported models were therefore estimated from 7043 datapoints (99% of cleaned total).

⁶ A reviewer pointed out that several of the related prime–target pairs were phonologically inconsistent (e.g. angelic-ANGEL). Three such items were identified in the transparent condition, eight in the opaque condition and nine in the form condition. Although inclusion of phonological consistency as a covariate did not significantly predict RT ($t = -0.85$, $p\text{MCMC} = 0.338$, raw RT Estimate = -7 ms) nor improve model fit ($\chi^2 = .07$, $p = 0.4$), it was retained in the model to adjust estimates of variance in the interactive terms.

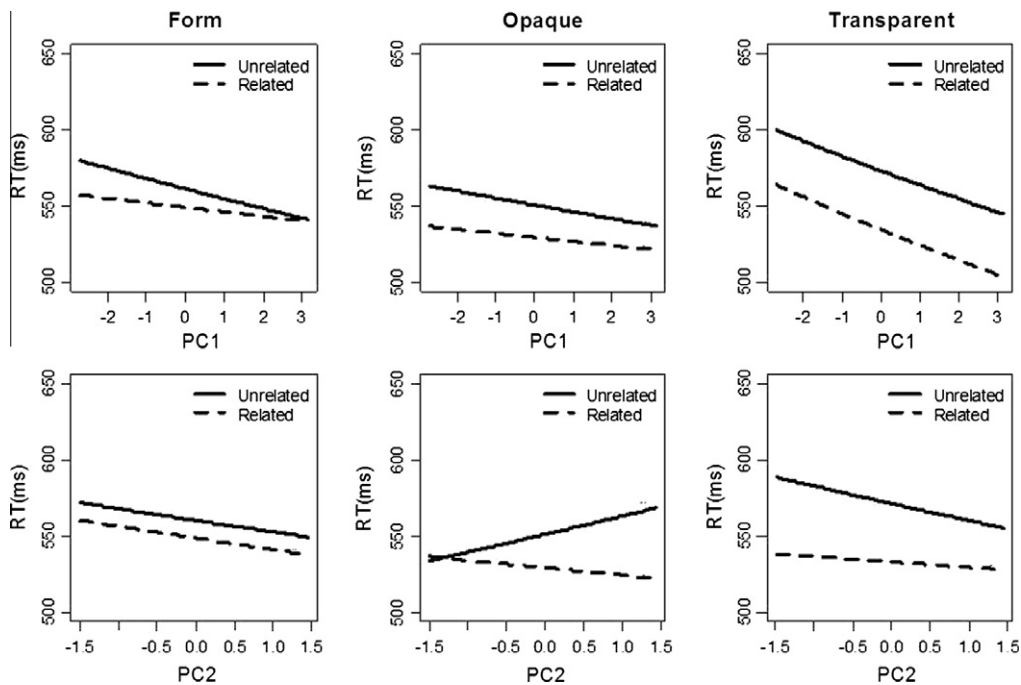


Fig. 1. LME model estimates of raw RT for Form, Opaque and Transparent stem targets preceded by Related or Unrelated primes as a function of individual differences on the first principal component (PC1; upper row) and second principal component (PC2; lower row). Higher scores on PC1 indicate higher overall proficiency. Higher scores on PC2 indicate that Spelling was higher than Vocabulary, while lower scores on PC2 indicate that Vocabulary was higher than spelling. (Model estimates of inverse RT have been converted to raw RT to facilitate interpretation).

vocabulary (high PC2) was associated with increased priming for opaque pairs, but reduced priming for transparent pairs. By contrast, higher vocabulary than spelling (low PC2) was associated with substantially stronger priming for transparent than opaque pairs. Similarly, the 3-way interaction of PC2 with the difference between opaque and form priming suggested that superior spelling was associated with greater facilitative priming for opaque pairs, but no change in priming for form pairs. Interestingly, this contrast was not significant in the models tested on inverse RT ($t = -1.66$, $p\text{MCMC} = 0.09$), but yielded a robust effect in the models tested on raw RT ($t = -2.48$, $p\text{MCMC} = 0.01$, Est. Raw RT = -32 ms). This suggests that the interaction may arise from differences in the tails of the RT distribution where the inverse transformation has its greatest impact (Kliegl, Masson, & Richter, 2010).

Thus, the LME analysis showed that priming did not vary with general proficiency but was systematically modulated by individual differences in relative performance on tests of spelling and vocabulary. Consistent with our predictions, superior spelling relative to vocabulary was associated with stronger priming for opaque than transparent pairs. By contrast, higher vocabulary than spelling was associated with increased priming for transparent pairs but minimal priming for opaque pairs. Based on Yap et al.'s (2009) evidence that individual differences in vocabulary modulate the time course of semantic priming across the RT distribution, we applied RT distributional analysis techniques to provide more refined evidence about the basis of the priming effects for transparent and opaque pairs and the manner in which they are modulated by individual differences.

RT distribution analyses

All correct and valid RTs for each prime–target condition were analyzed for each participant using the QMPE software (version 2.18; Cousineau, Brown, & Heathcote, 2004). Given the relatively limited number of datapoints for each condition, four quantile bands were calculated for each condition, rather than the 10 used by Yap et al. (2009). To investigate differences between the RT distributions for the different conditions in a non-parametric manner that does not depend on distributional assumptions (Andrews & Heathcote, 2001; Yap et al., 2009), we analyzed the quantile data for each condition.

Linear growth curve analysis of quantile estimates LME models were tested on the individual quantile estimates for each subject in each condition to track the time course of priming effects for the different target types and assess whether they are modulated by individual differences. These data are summarized in Fig. 2 which presents subject-averaged priming effects for each quantile for the whole sample. To assess differences in the time course of priming effects across conditions, growth curves of RT for each prime–target condition were modeled using a generalized matrix inversion of orthogonal polynomials up to the cubic trend.⁷ We only report effects involving the interactions with

⁷ In addition to including participants in the random structure of the model to take account of the significant variability between participants' intercept ($SD = 73$ ms), model fit was improved by also including the slope of participants' linear trend ($SD = 70$ ms, $\chi^2 = 717.82$, $p < .001$) and the correlation between intercept and slope ($r = 0.89$, $\chi^2 = 14.10$, $p < .001$, which indicated that faster participants tend to produce steeper increases in RT across quantiles).

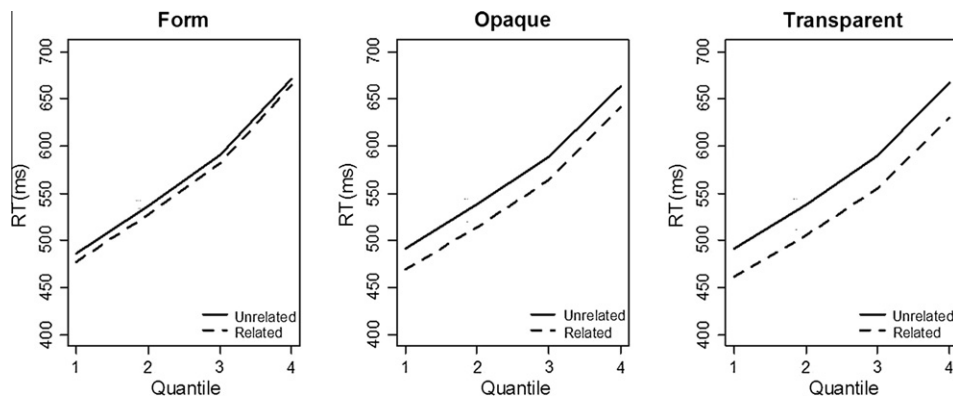


Fig. 2. Average lexical decision RT in each quantile for target words from the Form, Opaque and Transparent conditions, primed by Related and Unrelated primes.

these trends because the main effects and interactions of all the fixed factors exactly replicated the pattern of priming effects observed in the ANOVAs of raw RT and the interactions of priming with PC2, but not PC1, observed in the LME analyses.

With respect to the distributional locus of the priming effects, the average data illustrated in Fig. 2 show that the overall pattern of significantly less priming for form pairs than for the transparent and opaque conditions was evident across every quantile band (linear trend: $t = -1.23$, quadratic trend: $t = -0.39$, cubic trend: $t = -0.04$). This constant 'shift' in the RT distribution for related relative to unrelated pairs parallels the pattern observed in average semantic priming effects observed in lexical decision data for clearly presented targets (Balota et al., 2008; Yap et al., 2009). These distributional effects of priming were not significantly modulated by general proficiency as indexed by PC1 (all $t < 1.55$) but showed a number of significant interactions with PC2.

First, PC2 significantly interacted with the average linear trend in priming ($t = 2.82$, Estimate = 21 ms) across the RT distribution. To illustrate this effect, Fig. 3 presents the quantile data for related and unrelated pairs, averaged across the three target conditions, computed separately for participants categorized as High or Low on a mean split of PC2. These data show that the average priming effect decreased across the RT distribution for those of superior spelling but increased for those of superior vocabulary.

These differences in average priming were qualified by significant interactions between PC2 and the linear increase in priming for opaque pairs relative to both transparent pairs (Priming $\times$ Linear trend $\times$ PC2 $\times$ Transparent vs Opaque: $t = 3.55$, Estimate = 64 ms) and form pairs (Priming $\times$ Linear trend $\times$ PC2 $\times$ Opaque vs Form: $t = -5.02$, Estimate = -91 ms). To decompose these interactions, separate models assessing priming for each condition were run which each revealed significant interactions between PC2 and priming over quantiles,⁸ for different reasons. To summarize the different patterns of priming shown by high and low PC2 participants, Fig. 4 presents delta plots (Pratte, Rouder, Morey, &

Feng, 2010) of the priming effects over quantiles for each condition, separately for the two subgroups. The linear trend in priming significantly interacted with PC2 for form and transparent pairs (Priming $\times$ PC2 $\times$ Linear trend for Form: $t = 5.13$; Priming $\times$ PC2 $\times$ Linear trend for Transparent: $t = 3.01$) because in both conditions superior spelling relative to vocabulary (high PC2) was associated with a reduction in facilitative priming across quantiles, the opposite of the linear increase in facilitative priming for form and transparent pairs shown by low PC2 participants whose vocabulary is better than their spelling. For opaque pairs, priming was stronger for those with superior spelling than those with superior vocabulary (Priming $\times$ PC2: $t = -2.49$), and interacted significantly with the linear trend across quantiles (Priming $\times$ PC2 $\times$ Linear trend: $t = -2.85$), because priming for opaque pairs steadily increased across the RT distribution for those with superior spelling to vocabulary, but decreased for those with better vocabulary than spelling.

Thus, as shown in Fig. 4, both subgroups showed similar patterns of priming for the fastest quantile, which diverged for slower responses. Those with superior vocabulary relative to spelling, showed a selective increase in priming across the RT distribution for transparent stems but little priming for form or opaque pairs. However, in individuals with superior spelling, priming for opaque words showed a linear increase across the RT distribution while priming for form and transparent pairs tended to decrease.⁹

⁹ To address a reviewer's request that we complement our evidence of continuous effects of PC2 with a 'good old-fashioned factorial design' comparing participants categorized as high or low on PC2, we compared the priming effects on raw RT for the 15 (16%) highest and lowest scorers on PC2. These paralleled the LME analyses by showing that the High PC2 group, with superior spelling than vocabulary, showed stronger priming for opaque (47 ms) than transparent pairs (22 ms), and a small inhibitory priming effect for form pairs (-9 ms); while the superior vocabulary (Low PC2) group showed robust priming for transparent pairs (38 ms), no priming for opaque pairs (-2 ms) and an intermediate priming effect for form pairs (19 ms). Despite the small sample and lack of statistical control over item factors, ANOVA analysis showed a significant interaction between group and the difference between form and opaque pairs ($F(1,28) = 7.94$, $p < .01$) and a marginally significant interaction with the difference between opaque and transparent pairs ($F(1,28) = 3.74$, $p = .06$). The magnitude of these effects is compatible with the model estimates for the extreme ends of PC2 (see Fig. 1) showing that the differential patterns of priming are not carried by the most extreme individuals, an issue that was queried by another reviewer.

⁸ Including variability between participants in form, opaque or transparent priming in the random effect structure of each model (estimated at a standard deviation of 69 ms, 55 ms, and 58 ms, respectively) significantly improved model fit.

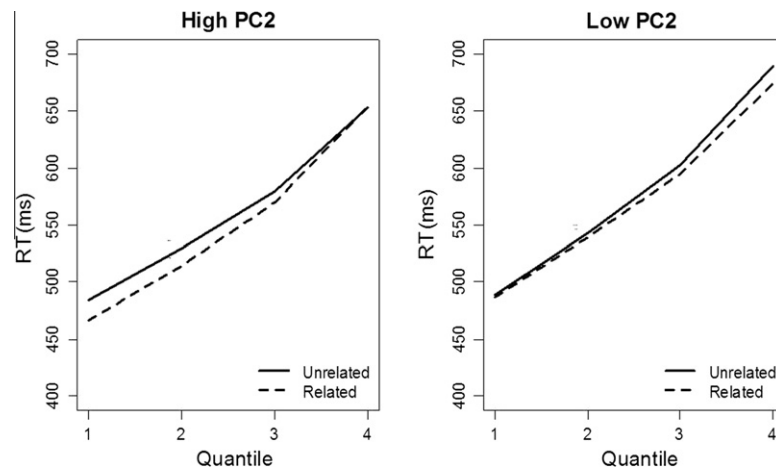


Fig. 3. RT for Related and Unrelated primes in each quantile, averaged over target condition, separately for participants above (High PC2) and below (Low PC2) the mean on the second principal component. Higher scores on PC2 indicate that Spelling was higher than Vocabulary, while lower scores on PC2 indicate that Vocabulary was higher than spelling.

In summary, the parallel shift patterns of priming observed for opaque and transparent pairs in the average RT quantile data obscure the contribution of individual differences in the relationship between spelling and vocabulary. Individuals with superior vocabulary than spelling ability showed a linear growth in priming for transparent relative to form and opaque pairs across the RT distribution, while those with superior spelling relative to vocabulary showed sustained priming effects for opaque pairs that are at least as strong as those for transparent pairs, particularly for slow RTs. The differential patterns of RT priming need to be interpreted cautiously because they are based on relatively small item numbers and the quantile analyses do not control for the item and trial-related factors included in the LME models tested on the full set of RT data.¹⁰ However, in combination with the interactions observed in the LME analysis of overall RT they provide novel insights into how individual differences modulate masked morphological priming.

General discussion

The present results provide new evidence about masked morphological priming that has important implications for

theories about the role of morphology in lexical retrieval. These theories have relied primarily on average performance for samples of skilled university student readers so we first consider the implications of our average data, which were based on larger samples and more sophisticated analysis procedures than most previous research on these issues, and then discuss how these effects were modulated by individual differences.

Morphological priming in the average data

Consistent with most previous literature, the results for the whole sample showed larger average priming effects for transparent (33 ms) than opaque (19 ms) or form (10 ms) pairs. The relative magnitude of the effects for transparent and opaque pairs is very similar to the average values of 30 ms and 23 ms, respectively, obtained across the 19 studies summarized by [Rastle and Davis \(2008\)](#). ANOVA analyses of the average mean RT data yielded statistically ambiguous outcomes because the intermediate priming effects obtained for opaque pairs did not differ significantly from the effects for either transparent or form pairs. However, clear outcomes emerged from the LME models which were tested on inverse RT, to normalize the distribution of scores, and controlled for a wide range of sources of variability between items and participants. These analyses showed a robust transparency effect, with significantly stronger priming for transparent than opaque pairs but no significant difference between opaque and form pairs. Interestingly, these outcomes precisely mirror the results for native English speakers reported by [Diependaele et al. \(2011\)](#) who also used a large sample of participants ($N = 65$) and conducted linear mixed effect analysis on inverse RT controlling for a range of stimulus attributes. Their estimates of the priming effects for transparent, opaque and form pairs were 36 ms, 15 ms and 1 ms, respectively, and paralleling the present analyses, the priming effect for opaque pairs did not significantly exceed the form priming effect.

Inspired by recent applications of RT distribution analyses to evaluate the locus of priming effects (e.g., [Balota](#)

¹⁰ The present design precludes LME models that directly parallel the quantile analyses by assessing the interactions between individual differences in PC2 and the change in priming with response speed across different prime conditions, because PC2 is a between-subjects variable and target type is a between-items variable. However, subsidiary LME analyses, using the same control variables as the primary analysis, assessed the random correlation between priming and response speed for just the Transparent condition, which showed the most marked difference between High and Low PC2 subgroups in the delta plots in [Fig. 4](#). These analyses confirmed that PC2 remained a strong predictor of variability in Transparent priming when item and trial factors were controlled and separate models tested on participants above (High) and below (Low) the mean on PC2 confirmed the quantile analysis by showing a marked increase in the Transparency effect with response speed for Low PC2 participants that was completely absent in the High PC2 subgroup. Despite the problems associated with directly comparing LME and quantile analyses, in response to a reviewer's request, we confirmed that the R^2 between the fitted values and the observed RTs was higher for the quantile analysis (0.952) than the LME analysis (0.501).

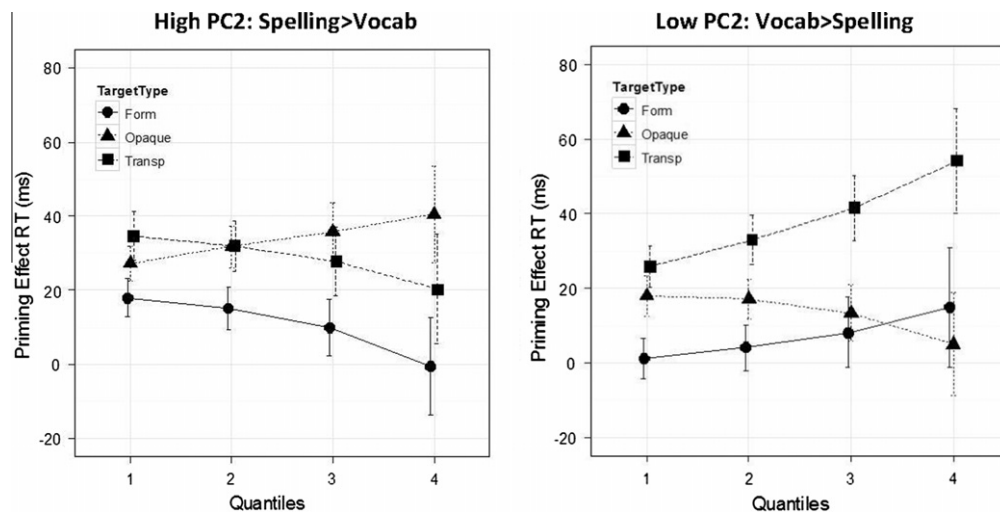


Fig. 4. Priming effects for each quantile, computed by subtracting the quantile mean for Related primes from that for Unrelated primes, separately for Transparent, Opaque and Form pairs. Participants above the mean on PC2, indicating higher scores on spelling than vocabulary, are presented in the left panel; and participants below the PC2 mean, indicating higher scores on vocabulary than spelling, are presented in the right panel.

et al., 2008; Johnson, Staub, & Fleri, 2012; Yap, Balota, Sibley, & Ratcliff, 2012; Yap et al., 2009) we sought further insight into the basis of the differential priming effects by analyzing their time course across the RT distribution. However, the average quantile data provided no evidence of differences in the pattern of priming for transparent and opaque pairs. The facilitation from related relative to unrelated primes in both conditions was statistically constant across the whole RT distribution (see Fig. 2).

This is the pattern of priming that has generally been observed in average RT data for both visible and masked semantic primes preceding clearly presented targets (Balota et al., 2008). Balota et al. (2008) argued that this generalized ‘shift’ pattern of priming is conceptually compatible with a modular headstart mechanism in which primes pre-activate relevant target words. Headstart accounts are compatible with parallel activation frameworks, such as Morton’s (1969) seminal logogen model, as well as serial accounts of lexical retrieval such as the entry-opening model (Forster & Davis, 1984) which also attributes priming effects to ‘savings’ in access time for targets preceded by related primes. However, distributional shifting can also be produced by adjusting the response criteria in a random walk model (Spieler, Balota, & Faust, 2000; Yap, Balota, Cortese, & Watson, 2006) and Yap et al.’s (2012) recent simulations of such shifts in semantic priming data using the diffusion model (Ratcliff, Gomez, & McKoon, 2004) attributed them to ‘non-decision processes’ such as encoding and response execution. This pattern of priming is therefore compatible with a range of theoretical mechanisms and models.

Thus, even with a much larger sample than usually assessed in studies of morphological priming, our average data cannot conclusively distinguish between different theoretical accounts. Our individual differences approach provides critical, novel evidence that explains the contradictions observed in published average data and provides new evidence that contributes to distinguishing the relative validity of theories of morphological priming.

Individual differences in morphological priming

Individual differences in general proficiency, indexed by the average of vocabulary and spelling, were associated with faster RT to transparent stems, regardless of whether primes were related or unrelated but did not significantly interact with priming in any of the analyses. Priming was, however, systematically modulated by PC2, which indexed differences between relative vocabulary and spelling ability, partialling out general proficiency. Critically, this dimension of individual differences was associated with systematic differences in sensitivity to orthographic and semantic components of morphological priming.

Consistent with the prediction that variance uniquely associated with vocabulary would tap semantic contributions to morphological priming, the LME analyses of inverse RT showed that individuals with the ‘semantic profile’¹¹ of higher vocabulary than spelling ability showed strong priming for transparent pairs but minimal priming for opaque pairs. However, individuals with the ‘orthographic profile’ of higher spelling than vocabulary, a combination that Andrews and Hersch (2010) identified with precise orthographic representations of words, showed at least as strong priming for opaque pairs as transparent pairs.

Further insight into how individual differences modulated priming was provided by the analyses of RT quantiles

¹¹ We will use the terms ‘orthographic’ and ‘semantic profile’ to refer to participants high and low on PC2, respectively, for convenience of expression. However, it is important to emphasise that the significant interactions observed in the analyses of inverse RT and RT quantiles used a continuous measure of PC2, rather than a categorical group factor. The fact that the general proficiency measure, PC1, accounted for 84% of total variability associated with spelling and vocabulary indicated that most participants tended to be good or bad at both spelling and vocabulary. Thus, the different patterns of priming associated with high and low scores on PC2 should not be taken to imply that there are two distinct kinds of readers. Rather, the significant interactions of priming with the continuous PC2 dimension, over and above individual differences in general proficiency, shows that a participant’s *relative* strength in spelling or vocabulary ability uniquely predicts systematic variations in the pattern of priming.

which showed that the constant magnitude of opaque and transparent priming across the RT distribution observed for the total sample arose from averaging over subgroups of participants who differed in relative sensitivity to orthographic and semantic similarity (see Figs. 3 and 4). The semantic profile was associated with a gradual increase in average priming across the RT distribution that was almost entirely due to a selective increase in priming for transparent pairs as RT lengthened. In complete contrast, the average priming effect associated with the orthographic profile decreased across the RT distribution for transparent and form but not opaque pairs.

The quantile data indicate that individual differences in the pattern of priming across conditions are more marked for slow than fast responses. Both profiles were associated with slightly stronger priming for transparent and opaque than form pairs for fast responses, but form priming was stronger for those with superior spelling, contributing to the significantly stronger average priming shown by this subgroup in the early quantiles. More marked differences in the pattern of priming across conditions emerged in slower RTs. It is tempting to attribute these patterns to individual differences in the relative time course of orthographic and semantic processing, paralleling recent claims that variations in the magnitude of the spatial Simon effect across the RT distribution “reflect the time course of activation of the ...[required] response” (Proctor, Miles, & Baroni, 2011, p. 263). However, such an interpretation may not be warranted because, in contrast to the Simon task, different items give rise to fast and slow lexical decision responses. Specifically, slower ‘word’ responses are usually associated with more difficult items – typically, longer, lower frequency words. This implies that individual differences in morphological processing have a stronger impact on retrieval of low frequency words, the items that show stronger morphological priming in average data (e.g., Burani & Carramazza, 1987). The novel evidence provided by the present research is that the morphological processing strategy applied to lower frequency words varies with individual differences in vocabulary and spelling.

These results extend previous evidence, reviewed in the Introduction, that individual differences modulate the early stages of lexical retrieval indexed by priming. Yap et al. (2009) attributed the ‘headstart’ priming shown by their high vocabulary participants for visible semantic primes to fluent, automatic processing of both high and low frequency words. The present results confirm the association between vocabulary and semantic priming, indexed here by stronger priming for transparent than opaque words. However, the more fine-grained probe provided by masked morphological primes, and the partitioning of individual differences in vocabulary and spelling, revealed that vocabulary captures only part of the systematic variance in early lexical retrieval. Consistent with our previous evidence that spelling ability predicts unique variance in masked orthographic priming (Andrews & Hersch, 2010; Andrews & Lo, 2012), higher spelling relative to vocabulary was associated with a distinct priming profile defined by strong priming for opaque pairs.

This pattern of individual differences provides strong constraints on theoretical accounts of masked morpholog-

ical priming by showing that a systematic component of individual differences amongst skilled readers appears to be explicable in terms of the balance between orthographic and semantic processing, assessed by independent measures of vocabulary and spelling obtained under quite different testing conditions than the experimental masked priming task. Although PC2 accounted for a smaller proportion of variance than the general proficiency indexed by PC1, its effects are important because of the highly systematic and theoretically plausible manner in which it modulated priming. It is also important to recognize that, because we computed orthogonal principal components, the differential strength of spelling and vocabulary indexed by PC2 is independent of individual differences in general proficiency. This point is methodologically critical because it indicates that efficient word identification is not, alone, sufficient to identify this source of individual difference. Theoretically, the independence of PC2 from general proficiency implies that the dimension of variability that is tapped by this factor relates to individual differences in the temporal dynamics of orthographic and semantic activation within the lexical network that are not captured by measures of the overall efficiency of lexical retrieval.

Implications for models of morphological priming

With respect to the ongoing debate about whether form is processed before meaning (e.g., Davis & Rastle, 2010; Feldman et al., 2009), the results suggest that the answer is ‘for some people’. The orthographic profile was associated with early form priming that reduced across the RT distribution but sustained priming for opaque pairs. This pattern appears broadly consistent with the predictions of morphographic models which assume that relatively little semantic information is extracted from a 50 ms masked prime (e.g., Davis & Rastle, 2010; Rastle & Davis, 2008). This model’s assumption that morphographic decomposition is an essential prerequisite for lexical retrieval is also consistent with the finding that the orthographic efficiency, indexed by superior spelling, predicts this pattern of priming. However, it is unclear how a pure morphographic model would explain the early divergence in patterns of priming for opaque relative to transparent and form pairs suggested by the quantile data, given that they were all equivalently orthographically similar. This feature of the data for superior spellers may be better accommodated by Taft and Nguyen-Hoan’s (2010) lemma model which allows for competition between the lemmas for morphologically-related words. Efficient orthographic processing indexed by superior spelling may support fast morphographic decomposition and lemma activation, leading to greater sensitivity to early competition between the lemmas for strongly related morphemes, like those for transparent words. However, form-first models cannot explain the pattern of priming shown by individuals higher in vocabulary than spelling. Relatively inefficient orthographic processing should impede activation of semantic information and therefore be associated with reduced or delayed priming for all conditions, rather than the selective growth of priming for transparent pairs associated with the semantic profile.

The systematic individual differences are more easily conceptualized within dual route frameworks which explicitly assume that different strategies for lexical retrieval operate in parallel and avoid the constraints imposed by assuming strict sequentiality of processing stages (e.g., Diependaele et al., 2011). Most previous evaluations of dual route models have focused on item factors, such as word frequency (e.g., Burani & Carramazza, 1987), or on changes in route emphasis in response to task demands (e.g., Andrews, 1986). In principle, there could also be individual differences in habitual patterns of reliance on different processing routes.

Recent evidence of complex interactions between morphological predictors in the skilled readers' average eye movement records for complex words (Kuperman, Schreuder, Bertram, & Baayen, 2009) has been interpreted as showing that morphemic and whole word routes to lexical access cannot be independent, as assumed by earlier dual route models (e.g., Schreuder & Baayen, 1995). Kuperman et al. (2009) therefore proposed a 'multiple-route model' based on "maximization of all opportunities – morphological, orthographic, phonological and contextual – for comprehension of the visual input" and argued that "models of morphological processing ... need to allow for a simultaneous processing of information at different levels without requiring strict sequentiality of processing stages" (Kuperman et al., 2009, p. 887). The present evidence that individual differences modulate the pattern of morphological processing raises the alternative possibility that different subgroups of readers adopt a different sequence of stages, or rely on a differently structured body of lexical knowledge.

In principle, such a possibility is consistent with PDP models which attribute individual differences to variations in the 'division of labor' between different components of the lexical network. Harm and Seidenberg (2004) showed that a PDP network trained to compute the meanings of printed monosyllabic words changed the weighting of different 'pathways' from orthography to meaning over the course of training. Early in training, accurate performance depended principally on connections from orthography to phonology to semantics ($O \rightarrow P \rightarrow S$) rather than on direct connections from orthography to semantics ($O \rightarrow S$) because the high correlations between orthography and phonology made it easier to learn $O \rightarrow P \rightarrow S$ connections than arbitrary $O \rightarrow S$ mappings. However, by the end of training, direct $O \rightarrow S$ connections made at least as strong a contribution to accurate computation of meaning as the mediated $O \rightarrow P \rightarrow S$ pathways because they allowed faster activation of meaning and contributed to the resolution of ambiguous forms such as homographs and homophones. Importantly in the present context, Harm and Seidenberg's (2004) model also developed sensitivity to morphological units such as past-tense *-ed* which was shown, by 'lesioning' the network, to be principally due to direct $O \rightarrow S$ connections. These connections also made a stronger contribution to the advantage for high over low frequency words than the $O \rightarrow P \rightarrow S$ connections, paralleling the role of word frequency in determining the division of labor between different pathways in dual route models. Similarly, individual differences might arise from

variations in the extent to which readers acquire direct $O \rightarrow S$ connections and, therefore, their sensitivity to morphological relationships.

Accounting for individual differences in terms of variability in the division of labor between different components of a PDP network is conceptually equivalent to localist multiple route frameworks that allow interactions between routes (e.g., Kuperman et al., 2009). However, the learning assumptions of the PDP framework allow for the possibility that individuals differ not just in their relative reliance on different pathways during on-line processing but in the internal representations of lexical knowledge they acquire, leading to systematic individual differences in performance over and above variability due to item and task characteristics.

The critical theoretical question raised by the present data is whether the different patterns of priming associated with the semantic and orthographic profile can be explained by a single model. Multiple route frameworks describe a lexical system that might accommodate such variability, but a complete account needs to specify how individual differences in representations and/or processing emerge during literacy acquisition. In the final section, we discuss two frameworks that have the potential to shed light on this issue.

Modeling individual differences in the development of morphological processing

The most direct evidence of how differences in the internal representations of words, acquired through training, might account for individual differences in morphological priming is provided by Plaut and Gonnerman's (2000) PDP simulations of cross-language differences between 'morphologically rich' languages, like Hebrew in which most words are morphologically complex, and relatively 'morphologically impoverished' languages like English. In contrast to the graded effects of morphological transparency usually found in average data for English-speaking participants, Hebrew speakers typically show 'pure' morphological priming that is independent of semantic similarity (e.g., Frost, Deutsch, & Forster, 2000). This appears to challenge PDP models' assumption that morphological priming depends on the convergence of orthographic and semantic similarity. To investigate this issue, Plaut and Gonnerman (2000) compared morphological priming in PDP simulations trained on artificial vocabularies constructed to emulate differences between morphologically rich and impoverished languages. The same set of critical stimuli, constructed to represent a graded degree of semantic transparency, were embedded in vocabularies in which the remaining words were either dominantly transparent (the morphologically rich environment) or all opaque (the impoverished environment). Both simulations showed a graded effect of semantic transparency but opaque pairs only yielded priming in the morphologically rich vocabulary, apparently paralleling the evidence of 'pure' morphological priming in Hebrew (Frost et al., 2000). These simulations provide concrete evidence of how the internal representations learned by PDP models are shaped by the structure of the vocabulary they are exposed to,

such that a single system learns systematically different similarity patterns amongst hidden unit weights for the same set of orthographic word forms. This sensitivity to morphological structure emerges because “in learning to map orthography to semantics, the network has developed hidden representations that compromise between orthographic and semantic similarity structure” (Plaut & Gonnerman, 2000, p. 459).

Although Plaut and Gonnerman's (2000) simulations were motivated by cross-language differences in morphological richness, the general principles about morphological representation in PDP frameworks that they demonstrate are relevant to understanding individual differences in sensitivity to morphological structure among speakers of a single language. Morphological structure gradually emerges in hidden unit representations across the learning process. This learning will depend not only on whether a particular level of systematicity is *available* in the vocabulary, but also on whether the learner extracts and uses it. Differences between English readers in the extent to which they extract dimensions of orthographic and or semantic similarity between words could potentially yield differences in the extent of morphological structuring of lexical knowledge that parallel the differences observed in Plaut and Gonnerman's (2000) simulations.

While this analysis provides a plausible general account of how individual differences in morphological priming might be accommodated within PDP frameworks, the specific predictions that appear to follow from Plaut and Gonnerman's (2000) simulations are opposite to the present findings. The most obvious mapping of our data to the simulations is to identify the semantic profile with greater sensitivity to semantic and therefore morphological similarity – paralleling learners of morphologically rich languages – and the orthographic profile with a focus on form-based rather than semantic similarity that leads to reduced sensitivity to morphological structure. However, this mapping predicts that priming for opaque pairs should occur in those with superior vocabulary but not in those with superior spelling – the reverse of our finding that superior spelling was associated with sustained priming for opaque pairs that was absent in those of superior vocabulary.

The artificial vocabularies used in the Plaut and Gonnerman (2000) simulations were intended to demonstrate that the general principles of the PDP framework can accommodate differential sensitivity to morphological structure, rather than to provide an explicit account of existing empirical evidence. Alternative PDP implementations may yield different outcomes. The impact of morphological systematicity on priming may be different if the model is highly overtrained, to parallel our use of highly skilled readers; or if a more realistic vocabulary and training regime is used in which phonological-semantic connections are trained before exposing the network to orthographic forms (Harm & Seidenberg, 2004). The present data provide important constraints for further extensions of PDP models of morphological priming.

Baayen et al.'s (2011) symbolic NDL model provides an alternative way of conceptualizing how individual differ-

ences might influence the structure and dynamics of morphological processing. In this model, lexical knowledge is represented in a two-layer network of associations between orthographic units (letters and bigrams) and symbolic semantic units¹² for both root morphemes (e.g., *hand* and *hands* map to the same lexical meaning) and 'grammatical meanings' (e.g., *hands* but not *hand* is associated with a unit representing plural, comparative). A model in which lexical decision RT was simulated by a weighted function of the word's lexical and grammatical meaning, modulated by two parameters reflecting activation from the word's most highly activated orthographic competitors, produced a reasonably good fit with single word lexical decision data for both inflected and derived word from the English Lexicon Project (ELP) database (Balota et al., 2007). A parallel model trained with the transparent, opaque and form pairs from Rastle et al. (2004), which assigned complex opaque words their own meanings rather than the meaning of their pseudo-stem, produced significant, statistically equivalent priming for transparent and opaque pairs. Thus, like the PDP approach, the NDL model yields priming for opaque words without either morphographic parsing or explicit representation of word forms.

One feature of the present data that emerges naturally from the NDL approach is the significant relationship observed between general proficiency and faster responses to transparent word stems. The strong, consistent associations between the lexical and grammatical meanings of words from transparent morphological families facilitate learning of the stems of transparent words. Higher reading proficiency may enhance this benefit leading to faster retrieval of transparent stems, independently of priming.

The modifications made to the NDL framework to simulate Rastle et al.'s (2004) finding of equivalent priming for opaque and transparent words also provide an alternative way of thinking about how individual differences in spelling relative to vocabulary might modulate morphological priming. In contrast to the “fully decompositional” NDL model (Baayen et al., 2011, p. 463) used to simulate the ELP data for inflected and derived words, the variant of NDL that successfully simulated Rastle et al.'s (2004) evidence of statistically equivalent priming for transparent and opaque stems included unique symbolic ‘semantic’ units for the meanings of pseudo-complex opaque words like *corner*. This qualifies the conclusion drawn by Baayen et al. (2011) that “naïve discriminative learning succeeds in capturing important aspects of morphological productivity, without having to posit separate representations for complex words” (p. 465). They also acknowledged that the absence of such separate meaning representations for derived words with partially transparent meanings (e.g., *business* from *busy*) may be responsible for the fully decompositional NDL model's failure to accurately simu-

¹² Although Baayen et al. (2011) refer to this layer of their model as representing semantics, it does not capture semantic similarity relationships between words and, as they acknowledge, is therefore incapable of capturing a range of semantic effects. The units at this level could equally well be identified with the lemma units that mediate between form and semantics in other conceptual frameworks (e.g., Taft & Nguyen-Hoan, 2010).

late the empirical effects of surface and base frequency in the ELP data for derived words.

Such considerations suggest an alternative approach to explaining individual differences in the pattern of morphological priming. In contrast to the PDP framework's emphasis on the importance of discovering, rather than stipulating, representational structures (Plaut & McClelland, 2010), the inductive learning process that characterizes the NDL approach allows the inclusion of symbolic units for complex word forms, as well as root morphemes, and for the relative strength of the associations between the units for complex forms, stems and grammatical meanings to be shaped through a contextually-based learning process. Within this framework, different patterns of morphological priming might reflect individual differences in the extent to which skilled readers have established lexical representations for complex word forms that are separate from those for their base morphemic constituents.

The orthographic profile of better spelling than vocabulary may be associated with the more bottom-up reading strategy that we have identified with precise, high quality lexical representations (Andrews & Bond, 2009; Hersch & Andrews, 2012). This strategy may foster the extraction of word-specific representations of complex word forms and frequently occurring orthographic patterns such as common suffixes, regardless of their morphological status, accounting for the sustained priming for opaque words. The reduced priming for slower responses to form and transparent words associated with this profile may reflect the stronger lexical competition that we have shown to be associated with higher levels of spelling ability (Andrews & Hersch, 2010; Andrews & Lo, 2012). In contrast, the semantic profile may reflect a more contextually-driven reading strategy that supports development of a more 'decompositional' lexicon defined by strong associations between morphological and semantic relatives and reduced sensitivity to orthographic similarity.

The general principle that skilled readers might vary in their reliance on stored word forms vs decompositional 'parsing' strategies is consistent with Baayen and Milin's (2010) interpretation of the trade-off between effects of whole word frequency and number of morphemes on self-paced reading times in terms of individual differences in relative reliance on declarative and procedural memory (Ullman, 2007). It remains to seen whether the different patterns of average RT and priming shown by our subgroups can be simulated by a single NDL model that includes representations of full word forms and varies the training regime to modulate the relative weighting of morphemic and whole word units.

More generally, the assumption that lexical knowledge consists of learned associations between orthography and semantics, which is shared by the NDL and PDP frameworks, allows for the possibility that differences in reading instruction and experience will shape the organization of lexical knowledge and the extent to which it captures systematic morphological relationships. Perhaps the most important theoretical implication of the present data for such learning models is that it may be necessary to assume

functionally localist representations of whole word forms (Andrews, 2006; Bowers, 2009) to accommodate the orthographic profile associated with higher spelling than vocabulary ability. This may reflect a more general tendency for perceptual expertise to drive an increase in "conformity to specific structural forms" of knowledge representation (McClelland et al., 2010, p. 354).

Conclusions

The present results provide novel evidence about masked morphological priming with important implications for theories of lexical retrieval. Our average data provide some resolution to the previous contradictory empirical findings (e.g., Davis & Rastle, 2010; Feldman et al., 2009) by confirming that, when LME analysis is used to exert sophisticated control over extraneous variability between items and participants, priming for pseudo-morphologically related pairs does not significantly exceed that for orthographically matched controls (e.g., Diependaele, Duñabeitia, Morris, & Keuleers, 2011; Feldman et al., 2009). However, the results challenge the validity of relying solely on average data to construct and test theories of visual word recognition (Andrews, 2012) because the average effects of morphological priming obscured systematic individual differences. In contrast to the assumptions of form-first models, some skilled readers of English show graded effects of semantic transparency but no priming for opaque pairs, while others show equivalent priming in both conditions which is difficult to reconcile with models predicting graded effects of transparency.

Multiple route models (e.g., Kuperman et al., 2009) can attribute systematic individual differences to variations in the weighting of different routes or levels of processing. However, such accounts have little predictive utility without a clear specification of how different representations and processes develop through reading experience. PDP frameworks provide a principled account of how differences in readers' sensitivity to orthographic or semantic similarity could yield differences in morphological priming. However, the most relevant simulations reported so far (Plaut & Gonnerman, 2000) predict the opposite pattern of individual differences to that observed in our sample of skilled readers. The NDL model (Baayen et al., 2011) provides an alternative learning-based account that allows for the possibility that skilled readers vary in the extent to which they have acquired representations of complex word forms.

Such an approach is consistent with the lexical quality account of skilled reading which identifies skilled reading with precise orthographic representations of lexical forms that support rapid, context-free word identification (Andrews, 2012; Perfetti, 2007). This conclusion is consistent with studies of individual differences in more naturalistic sentence reading contexts in which efficient word identification is the best predictor of the eye movements of adult readers (Kuperman & Van Dyke, 2011) implying that more skilled readers rely more on bottom-up word identification than contextually-based prediction (Ashby, Rayner, & Clifton, 2005).

These findings converge with our previous research on masked orthographic priming (Andrews & Hersch, 2010; Andrews & Lo, 2012) in highlighting the limitations of relying solely on average RT data to develop and test models of visual word recognition and the important role that investigations of individual differences amongst skilled readers can play in refining theoretical understanding of lexical processing.

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Appendix A.

Prime and target word pairs used in the experiment.

Transparent			Opaque			Form		
Target	Related	Unrelated	Target	Related	Unrelated	Target	Related	Unrelated
DREAM	dreamer	masonry	FLICK	flicker	adviser	SIGH	sight	happy
REACT	reaction	physical	QUEST	question	actually	EXTRA	extract	justify
CREAM	creamy	watery	IRON	irony	sandy	INFER ^a	inferno	frilled
FLESH	fleshy	lovers	EARL	early	within	GLAD	glade	cuffs
DRUNK	drunkard	feathery	BUZZ	buzzard	loyally	SURF	surface	medical
ERUPT	eruption	vicarage	AUDIT	audition	selfless	PLAIN	plaintiff	absurdity
ALARM	alarming	composer	GRUEL	grueling	existent	SQUAW ^a	squawk	oddity
RENEW	renewable	exemption	AMEN	amenable	palpably	STUB	stubborn	moisture
GREEN	greenery	snobbish	SCULL	scullery	narrowly	STIR	stirrup	buoyant
BARON	baronet	voucher	TRUMP	trumpet	chatter	CANDID	candidacy	epileptic
TEACH	teacher	finally	COAST	coaster	muffler	PULP	pulpit	gifted
FLOAT	floater	missive	ARCH	archer	feudal	HEAVE	heaven	firmly
LEGEND	legendary	anxiously	SECRET	secretary	obviously	SHOVE	shovel	tricky
DIET	dietary	wearily	GLOSS	glossary	sufferer	FORCE	forceps	prudish
ADOPT	adopted	kingdom	CROOK	crooked	pottery	ELECT	electron	suburban
NYMPH	nymphet	acutely	BRISK	brisket	foundry	VILLA	villain	grossly
CRITIC	critical	tendency	RATION	rational	steadily	DIAL	dialogue	lateness
FILTH	filthy	harden	HEART	heartly	folder	STUN	stunt	misty
OXYGEN	oxygenate	fossilize	LIQUID	liquidate	extremism	WEIR ^a	weird	manly
BOMB	bomber	lessen	NUMB	number	really	STUD	studio	gently
GOVERN	government	situation	DEPART	department	production	BASIL	basilica	princely
TUFT ^a	tufted	silken	UNIT	united	others	QUART	quartz	roller
ANGEL	angelic	watcher	CRYPT	cryptic	dweller	JERK	jerkin	twisty
WIDOW	widowed	beastly	STILT	stilted	gaseous	TWIN	twinkle	cheaply
GUILT	guilty	formal	PLUCK	plucky	winger	GALA ^a	galaxy	keeper
VIEW	viewer	ranger	SNIP	sniper	hourly	PHONE	phonetic	dreadful
ACRE	acreage	plunder	PLUM	plumage	broiler	FREE	freeze	golden
BLOOD	bloody	active	CRAFT	crafty	vainly	BROTH	brothel	warfare
ARTIST	artistry	calmness	INFANT	infantry	validity	STAMP	stampede	defector
SCALD	scalding	jauntily	FLEET	fleeting	simplify	SCRAP	scrape	ninety

^a Denotes excluded targets.

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